

Mahek Patel

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EDUCATION

Bachelor of Technology in Computer Science and Engineering
KIET Group of Institutions, Ghaziabad

Expected 2029

TECHNICAL SKILLS

Languages	Python, C, JavaScript, SQL
Backend	FastAPI, REST APIs, WebSockets, AsyncIO, SQLAlchemy, Socket Programming
Frontend	React.js, HTML, CSS
Databases	PostgreSQL, MongoDB, SQLite
Tools	Git, GitHub, Linux, VS Code, Postman
AI/ML	LangChain, Hugging Face Transformers, OpenAI APIs, Prompt Engineering, RAG

PROJECTS

Socket Talk – Distributed Real-Time Messaging Platform

FastAPI, WebSockets, AsyncIO

- Built a real-time messaging platform using WebSockets for persistent low-latency communication.
- Used AsyncIO for concurrent client handling and scalable message broadcasting.
- Implemented connection lifecycle management and real-time synchronization.
- Improved throughput using FastAPI's asynchronous execution model.

TaskFlow API – Enterprise Task Management Backend

FastAPI, PostgreSQL

- Developed a RESTful backend with authentication, task management and pagination.
- Designed normalized PostgreSQL schemas using SQLAlchemy ORM.
- Built modular architecture with routers, services and dependency injection.
- Implemented validation and error handling for reliable APIs.

HTTP Proxy Server

Python, Socket Programming, HTTP

- Built a multithreaded HTTP forward proxy server using TCP sockets.
- Implemented HTTP request parsing, header forwarding and response relaying.
- Supported concurrent clients with robust connection and error handling.
- Followed HTTP protocol standards using modular networking components.

LLM Conversational Assistant

Python, Hugging Face, LangChain

- Built an AI assistant using transformer-based language models.
- Integrated Hugging Face pipelines for inference and context management.
- Applied prompt engineering to improve response quality.
- Designed modular pipelines ready for RAG integration.

OPEN SOURCE CONTRIBUTIONS

- Contributed to open-source projects through the Linux Foundation LFX Mentorship program.
- Submitted pull requests, addressed maintainer review comments, and improved documentation.
- Collaborated with maintainers through GitHub discussions and code reviews.
- GitHub: github.com/MahekPatel-2403

ACHIEVEMENTS

- Linux Foundation LFX open-source contributor.
- Active backend developer focused on distributed systems, APIs and AI applications.